

Variants of the leading theorem: $C \neq 0$ and nonnegative face amplitudes

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Abstract

Following the audit in the second Astra consult, the leading theorem is exposed in coefficient-based form: the asymptotic equivalence $Z \sim CN^{-p} \log^{d_0} N$ needs only $C \neq 0$, and $C > 0$ holds as soon as $\eta(0, \cdot) \geq 0$ on the noncritical box and $\eta(0, v_0) > 0$ at one interior point, since the positive kernel and weights make the coefficient a positive integral over an open set of positive measure contained in the support. Zero sorry/axiom.

1 The things formalised

```
theorem blockStateIntegral_isEquivalent_of_ne ... (hC : blockCoeff b p k k' h' 0) :
  (fun N : => blockStateIntegral b N k h k' h' ) ~[atTop]
  fun N : => blockCoeff b p k k' h' * (N ^ (-p) * Real.log N ^ d)
theorem integral_blockDivisorDensity_pos_of_nonneg ...
  (henn : v, ( i, 0 < v i v i b) → 0 e v)
  (v : Fin d' → ) (hv : i, 0 < v i v i < b) (hpos : 0 < e v) :
  0 < v, blockDivisorDensity p k d k' h' a e v (boxMeasure b d')
theorem blockStateIntegral_isEquivalent_of_nonneg ... :
  0 < blockCoeff b p k k' h'
  (fun N : => blockStateIntegral b N k h k' h' ) ~[atTop]
  fun N : => blockCoeff b p k k' h' * (N ^ (-p) * Real.log N ^ d)
```

2 Proof strategy

`integral_pos_iff_support_of_nonneg_ae` reduces positivity to a positive measure of the support; the open set $\{e > 0\} \cap (0, b)^{d'}$ contains v_0 , hence has positive Lebesgue measure (`IsOpen.measure_pos`), lies in the closed box, and lies in the support because every other factor of the density is positive there.

3 Roadblocks and resolutions

- `Set.mem_univ_pi` cannot be rewritten at a membership in an intersection; project the component first.

4 Outcome

The leading theorem now has the recommended interface: unconditional scaled limit, $C \neq 0$ equivalence, $C > 0$ under a mild sign condition. A third Astra consult on the all-order theory is in

progress.